

Microelectronic Circuits I

Cadence Virtuoso Intro.

Prof. Jeong Hoan Park

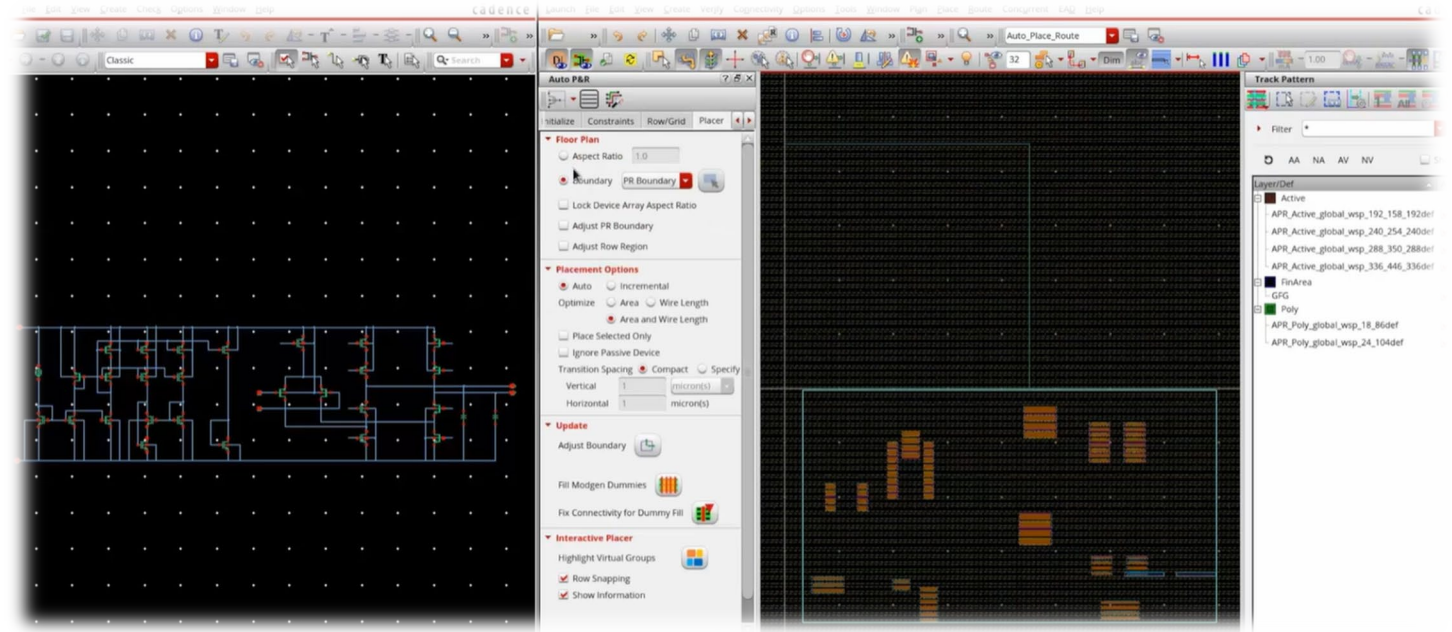
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Cadence Virtuoso

❑ Cadence Virtuoso® Platform

- Industry-standard environment for **custom analog** and mixed-signal IC design
- Supports **schematic**, layout, simulation, and verification in one platform
- High-accuracy simulation using **Spectre® (~SPICE)**
- Tight integration with foundry PDKs (DRC/LVS/RCX)
- Widely used in academia and industry tape-out flows



Example 1 – Resistive Circuit Simulation

E2.20 Find V_1 in Fig. E2.20.

Answer:

$$V_1 = 12 \text{ V.}$$

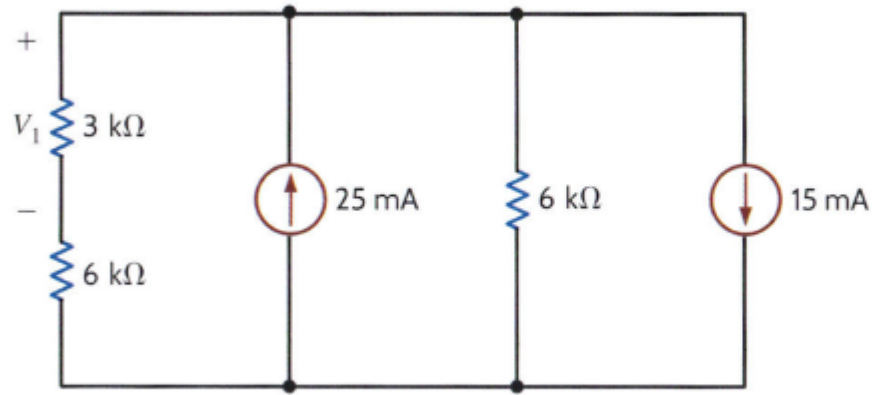
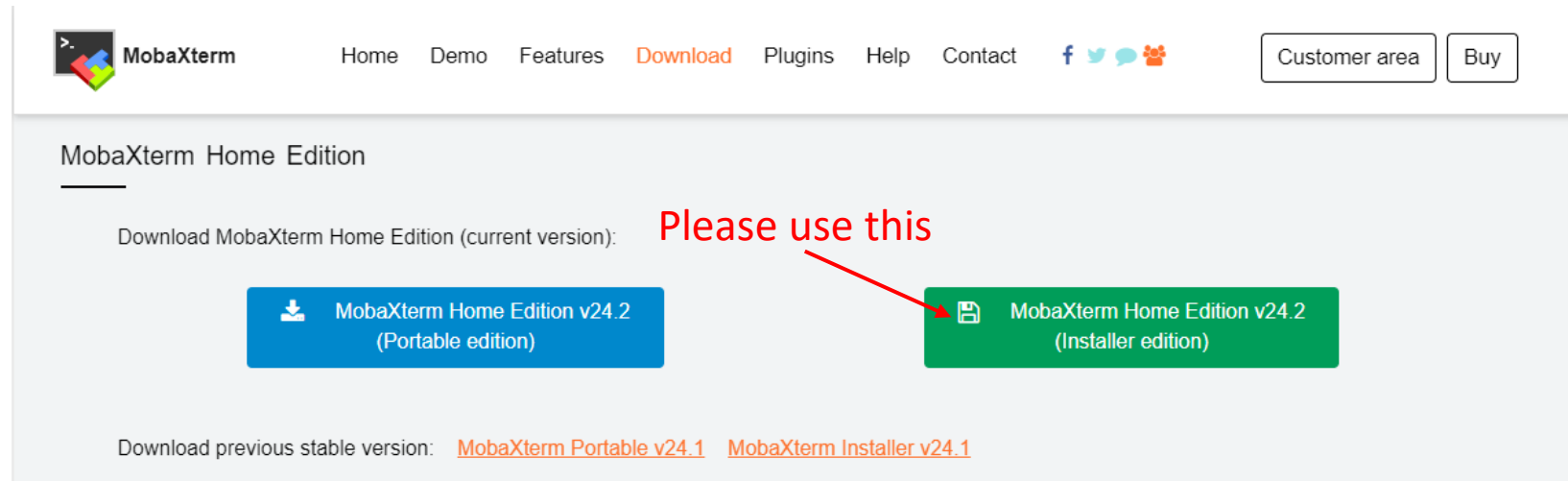


FIGURE E2.20

Installation of mobaxterm

❑ Sites

- <https://mobaxterm.mobatek.net/download.html>



Linux Basic Commands & Launch Virtuoso

❑ mkdir/cd/pwd/ls

- mkdir (make directory): making folder with specific name
- cd (change directory):
- pwd (present work directory):
- ls (list): list all files in present work directory
- Example

```
-[id@dice]$ mkdir me1
-[id@dice]$ ls
-[id@dice]$ rmdir me1
-[id@dice]$ ls -a
-[id@dice]$ mkdir ~/me1/schem
-[id@dice]$ cd project
-[id@dice]$ cd ~/me1/schem
-[id@dice]$ virtuoso &
```

```
• MobaXterm Personal Edition v24.2 •
(SSH client, X server and network tools)

► SSH session to jhpark@163.180.160.78
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✓ (remote display is forwarded through SSH)

► For more info, ctrl+click on help or visit our website.

Last login: Fri Aug 30 05:04:45 2024 from 163.180.159.157
[jhpark@dice ~]$ mkdir project
[jhpark@dice ~]$ rmdir project/
[jhpark@dice ~]$ mkdir project
[jhpark@dice ~]$ ls
a Desktop dice Documents Downloads eda.sh Music Pictures project Public soc Templates Videos
[jhpark@dice ~]$
```

Optional - Linux Basic Commands

❑ cp/mv/rm/cat

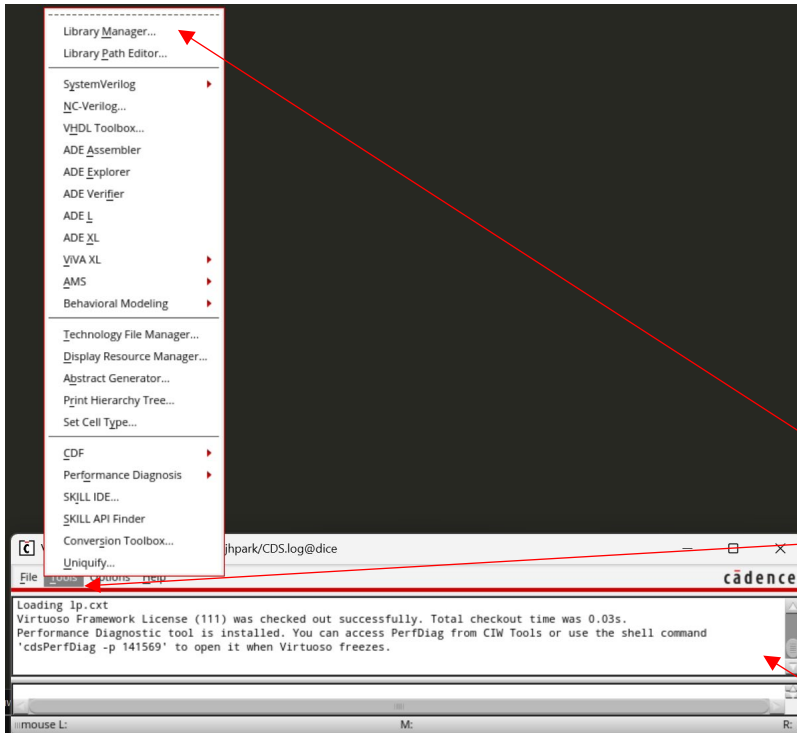
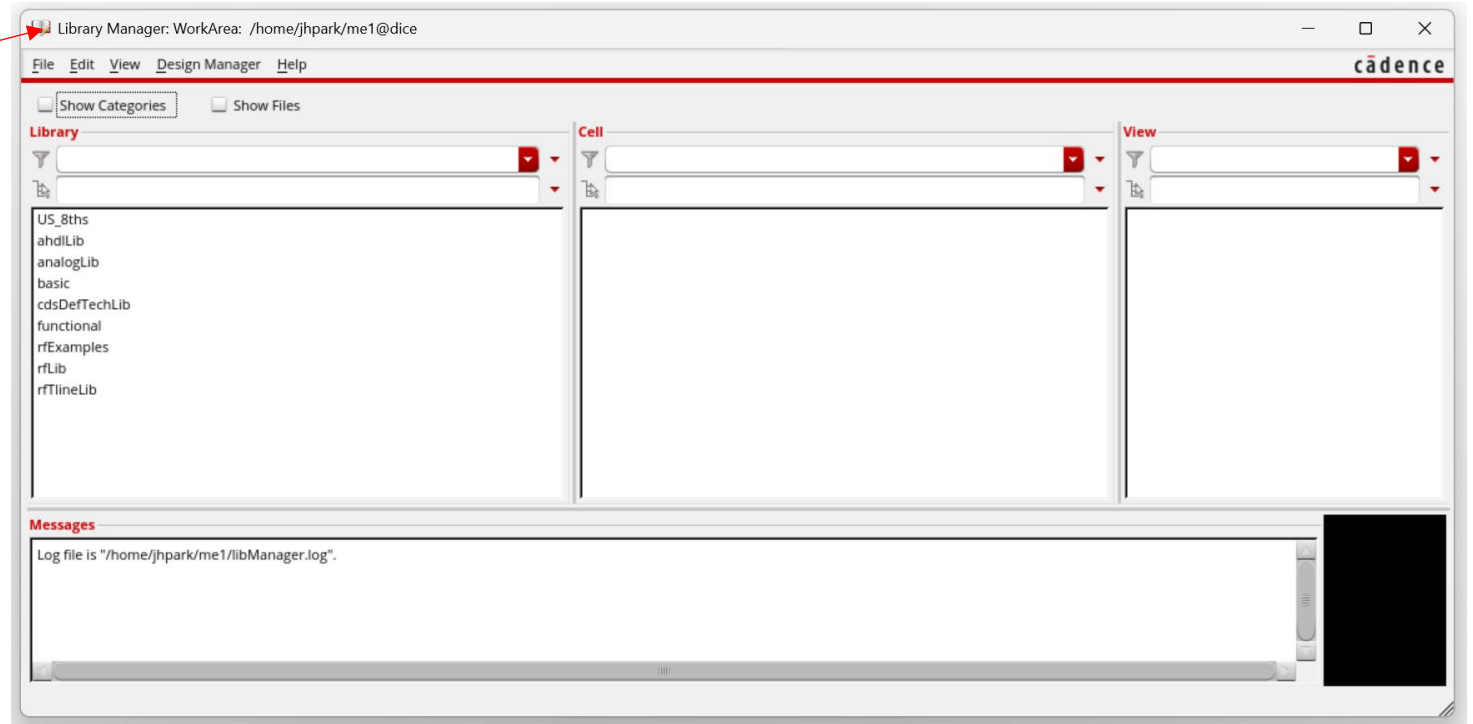
- cp (copy): cp [option] SOURCE DEST
 - Useful option: -r, -p, -v
- mv (move): mv [option] SOURCE DEST
- rm (remove): rm [option] FILES...
 - Useful option: -r, -f
- cat (concatentate) : concatenate FILE1 FILE2 ...

❑ Examples

- [id@dice]\$ ll > file.list
- [id@dice]\$ cp file.list ./rtl
- [id@dice]\$ cd rtl
- [id@dice]\$ ls
- [id@dice]\$ mv ./file.list ../../sim
- [id@dice]\$ ls
- [id@dice]\$ cd ../../sim
- [id@dice]\$ ls
- [id@dice]\$ cat file.list
- [id@dice]\$ gedit file.list
- [id@dice]\$ rm file.list

Create Lib.

This is Library Manager



2. Click Library Manager

1. Click Tools

We call that this is CIW

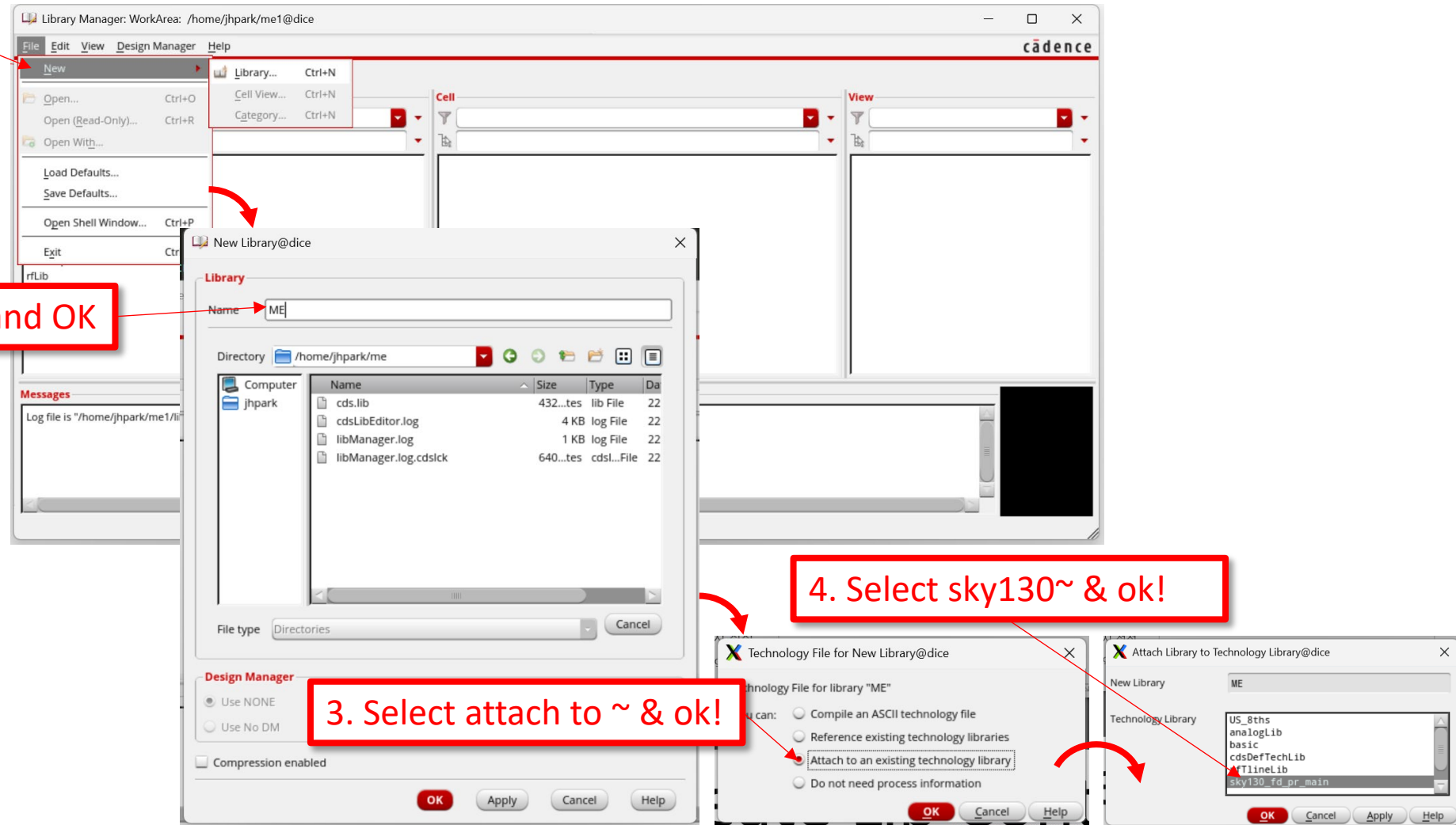
Create Lib Cont'

1. Click File → New → Library

2. Type "ME" and OK

3. Select attach to ~ & ok!

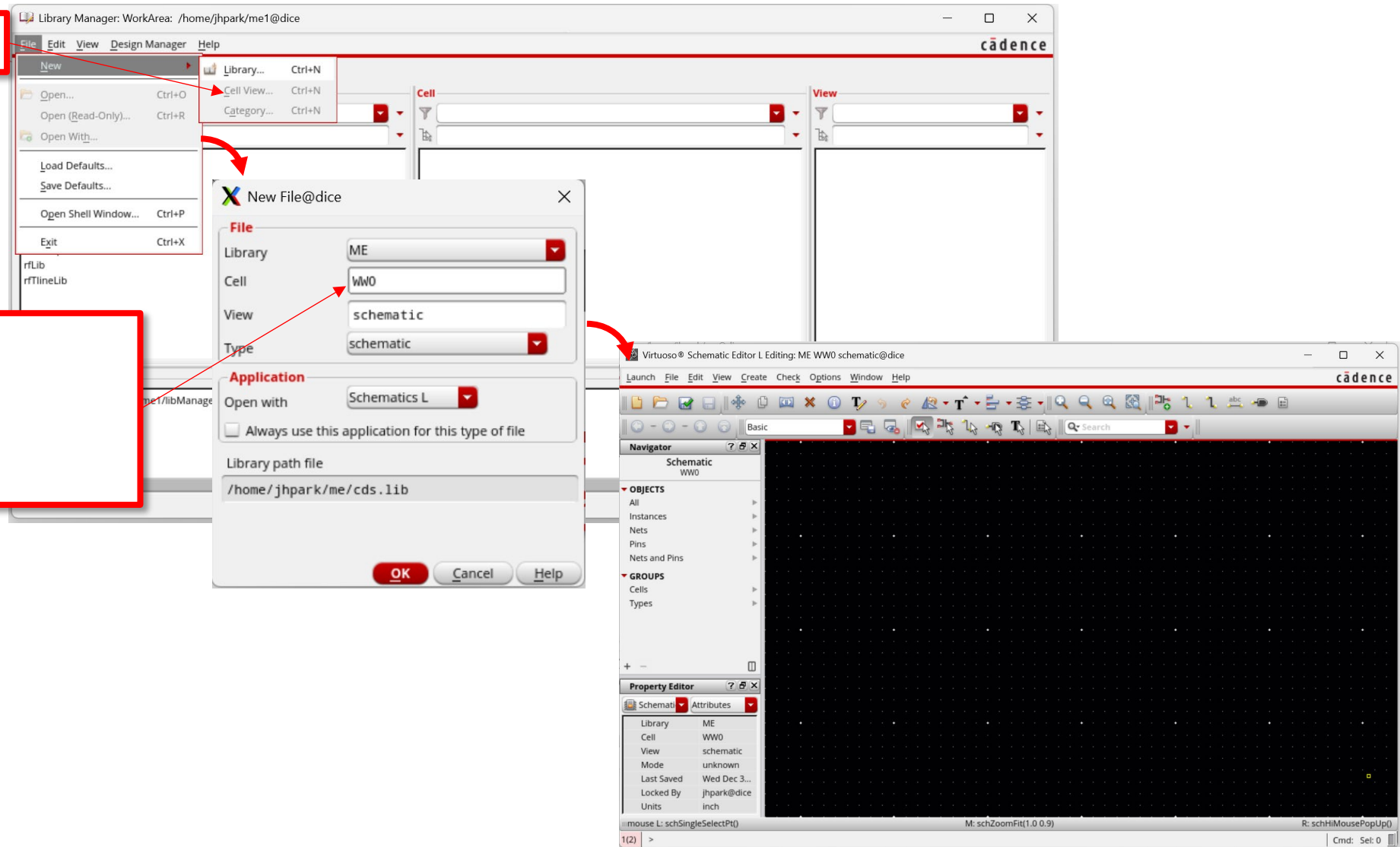
4. Select sky130~ & ok!



Create Cells

1. Click File → New → Cell View

2. Lib: ME
→ Cell: WW0
→ View: schematic
→ OK



Schematic View

Schematic Basic Commands

□ Basic commands

Shortcut	Description	Detailed Explanation
i	Insert	Insert an instance from the selected library and cell into the schematic.
c	Copy	Copy selected objects such as instances, wires, or labels.
m	Move	Move selected objects while preserving connectivity.
s	Stretch	Stretch wires or adjust pin positions without breaking connections.
l	Add Label	Add a net label. Nets with the same label name are electrically connected, even if not physically wired.
p	Add Port	Add a port for hierarchical connectivity (input/output/inout). Used when creating symbols.
q	Property	Open the property editor to modify parameters (e.g., W/L, instance name, net name).
u	Undo	Undo the previous operation.
x	Check and Save	Perform a schematic check for errors (floating nets, invalid connections) and save the design.
o	Options	Open editor options, including grid spacing, snap settings, and display options.

Add Instance - Resistor

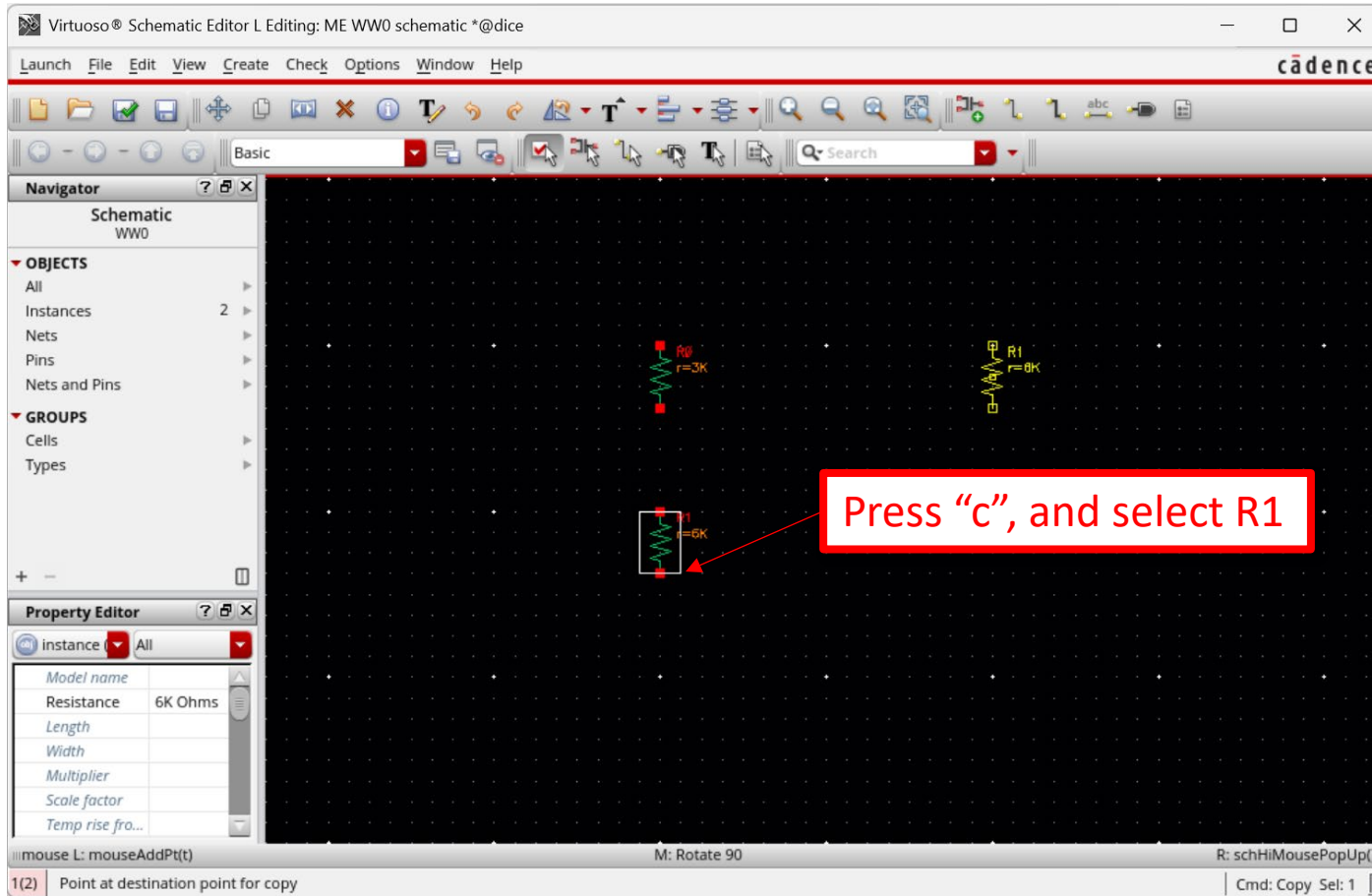
Lib: analogLib
→ Cell: res
→ View: symbol
→ Click in schematic!

Change the property

Once a resistor is selected, press "q"

Apply To: only current instance
Show: ☐ system ☒ user ☒ CDF
Browse Reset Instance Labels Display
Value Display
analogLib off
res off
symbol off
Instance Name R0 off
Add Delete Modify
CDF Parameter Value Display
Model name off
Resistance 1K Ohms off
Length off
Width off
Multiplier off
Scale factor off
Temp rise from ambient off
Temperature coefficient 1 off
Temperature coefficient 2 off
Resistance Form off
Generate noise? ☒ off
Capacitance off
Alias for Lin. temp. co. off
Alias for Quad temp. co. off
Lin temp co of lin cap off
Quad temp co of lin cap off
Resistance Scaling Factor off
Capacitance Scaling Factor off
AC resistance off
Temperature difference off
OK Cancel Apply Defaults Previous Next Help

Copy Instance - Resistor



Single-Time-Constant Networks Simulation

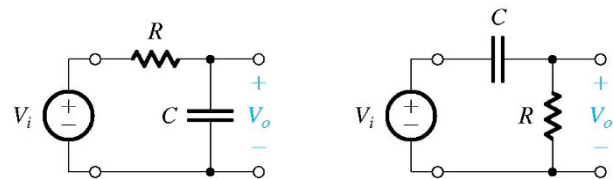
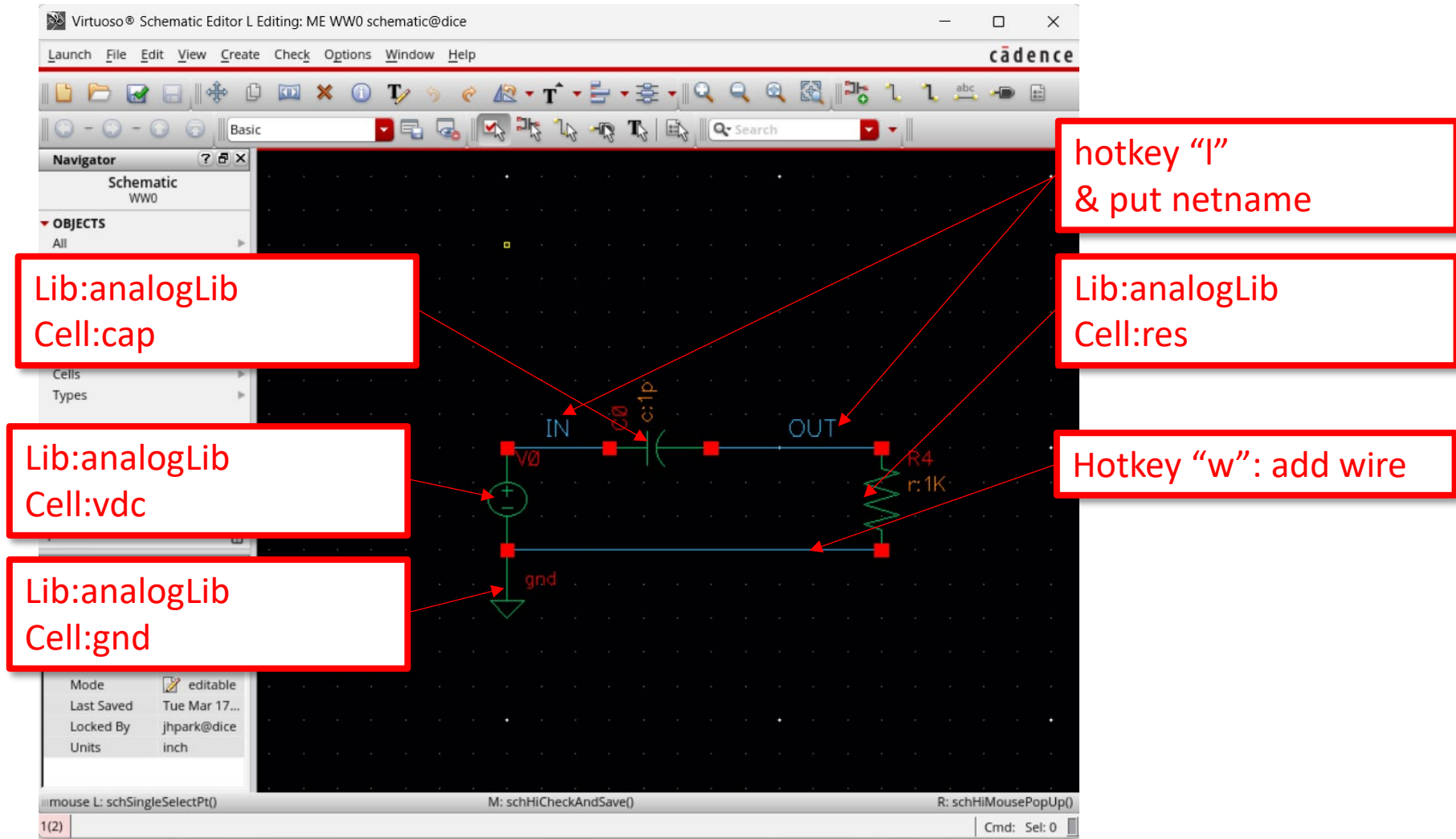


Table 1.2 Frequency Response of STC Networks

	Low-Pass (LP)	High-Pass (HP)
Transfer Function $T(s)$	$\frac{K}{1 + (s/\omega_0)}$	$\frac{Ks}{s + \omega_0}$
Transfer Function (for physical frequencies) $T(j\omega)$	$\frac{K}{1 + j(\omega/\omega_0)}$	$\frac{K}{1 - j(\omega_0/\omega)}$
Magnitude Response $ T(j\omega) $	$\frac{ K }{\sqrt{1 + (\omega/\omega_0)^2}}$	$\frac{ K }{\sqrt{1 + (\omega_0/\omega)^2}}$
Phase Response $\angle T(j\omega)$	$-\tan^{-1}(\omega/\omega_0)$	$\tan^{-1}(\omega_0/\omega)$
Transmission at $\omega = 0$ (dc)	K	0
Transmission at $\omega = \infty$	0	K
3-dB Frequency	$\omega_0 = 1/\tau$; $\tau \equiv$ time constant $\tau = CR$ or L/R	
Bode Plots	in Fig. 1.23	in Fig. 1.24

Configure Schematic



Configure Schematic

Edit Object Properties@dice

Apply To: ☒ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	☐ off
Cell Name	vdc	☐ off
View Name	symbol1	☐ off
Instance Name	V0	☐ off

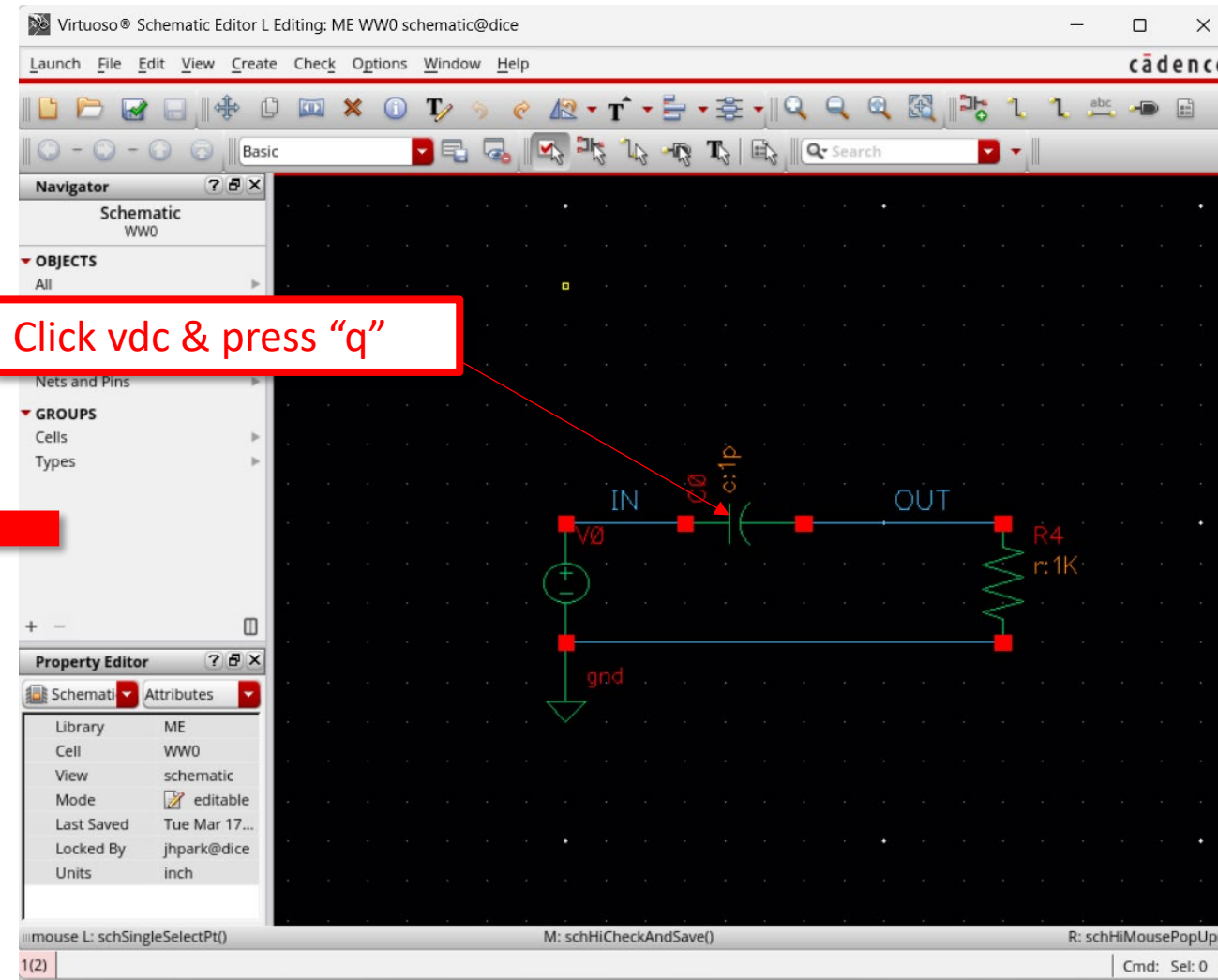
Add Delete Modify

User Property	Master Value	Local Value	Display
Ivsignore	TRUE		☐ off

CDF Parameter

Parameter	Value	Display
Noise file name		☐ off
Number of noise/freq pairs	0	☐ off
DC voltage	1 V	☐ off
AC magnitude	1 V	☐ off
AC phase	0	☐ off
XF magnitude		☐ off
PAC magnitude		☐ off
PAC phase		☐ off
Temperature coefficient 1		☐ off
Temperature coefficient 2		☐ off
Nominal temperature		☐ off

OK Cancel Apply Defaults Previous Next Help

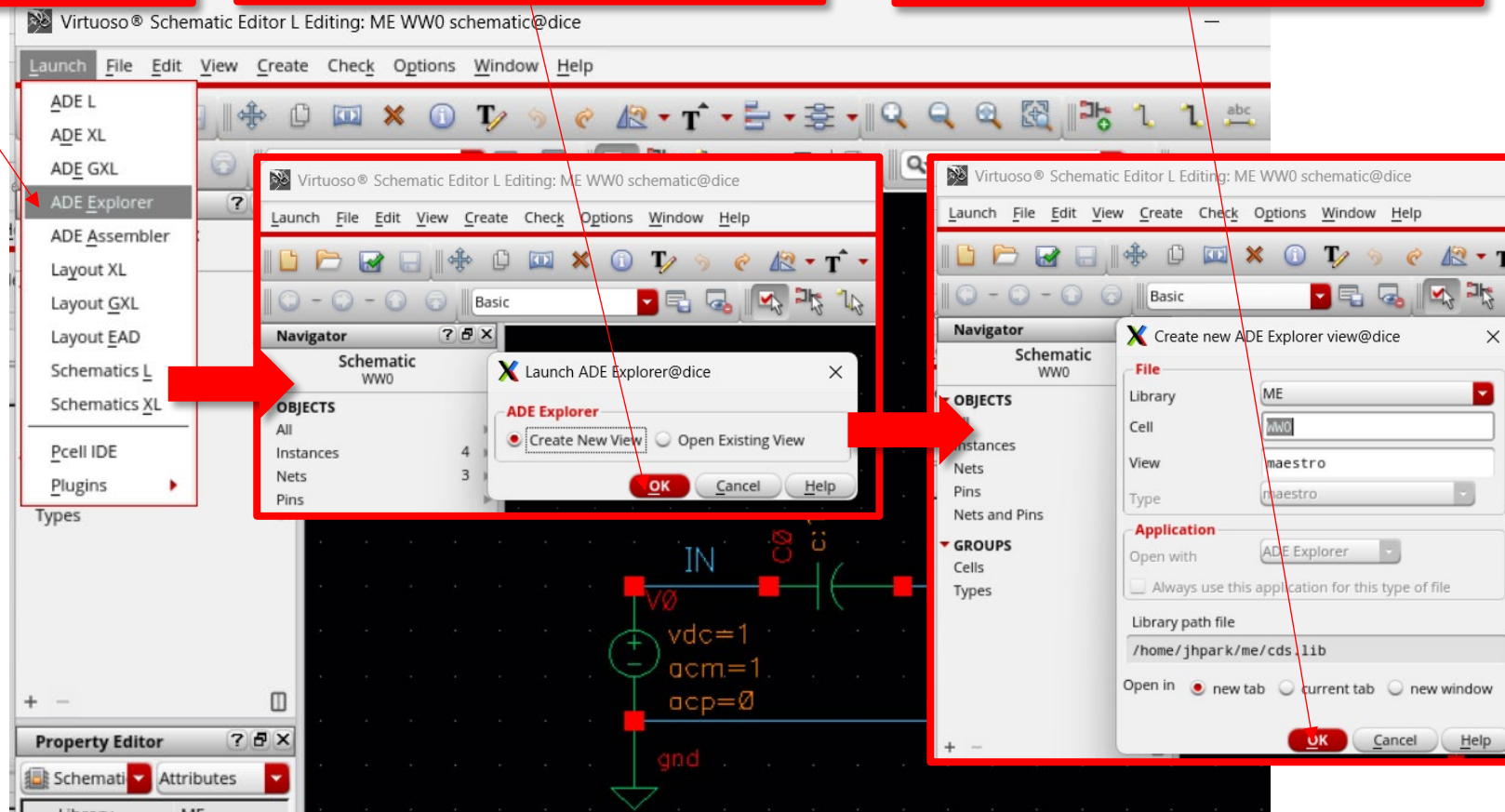


Launch ADE Explorer

1. Click launch/ ADE Explorer

2. Choose "Create New View"

2. click "Ok"



ADE Explorer - analysis

The screenshot displays the Cadence Virtuoso ADE Explorer interface. The main window is titled "Virtuoso® ADE Explorer Editing: ME_WW0 maestro@dice". The left sidebar shows a tree view with "ME_WW0_1" selected, containing "Simulator: spectre", "Analyses", "Design Variables", "Parameters", "Corners", "Reliability Analyses", "Monte Carlo Sampling", and "Checks/Asserts". The "Analyses" section is expanded, showing a list of analyses. A red box labeled "Analysis 'ac', 'tran'" points to the "Analyses" section. Two dialog boxes are open: "Choosing Analyses -- ADE Explorer@dice" and "Choosing Analyses -- ADE Explorer@dice". The first dialog shows the "Transient Analysis" section with "Stop Time" set to "1m" (highlighted by a red box labeled "1ms"). The second dialog shows the "AC Analysis" section with "Sweep Variable" set to "Frequency" and "Sweep Range" set to "Start-Stop" with "Start" at "0.1" and "Stop" at "1G" (highlighted by a red box labeled "0.1 ~ 1GHz").

Analysis "ac", "tran"

1ms

0.1 ~ 1GHz

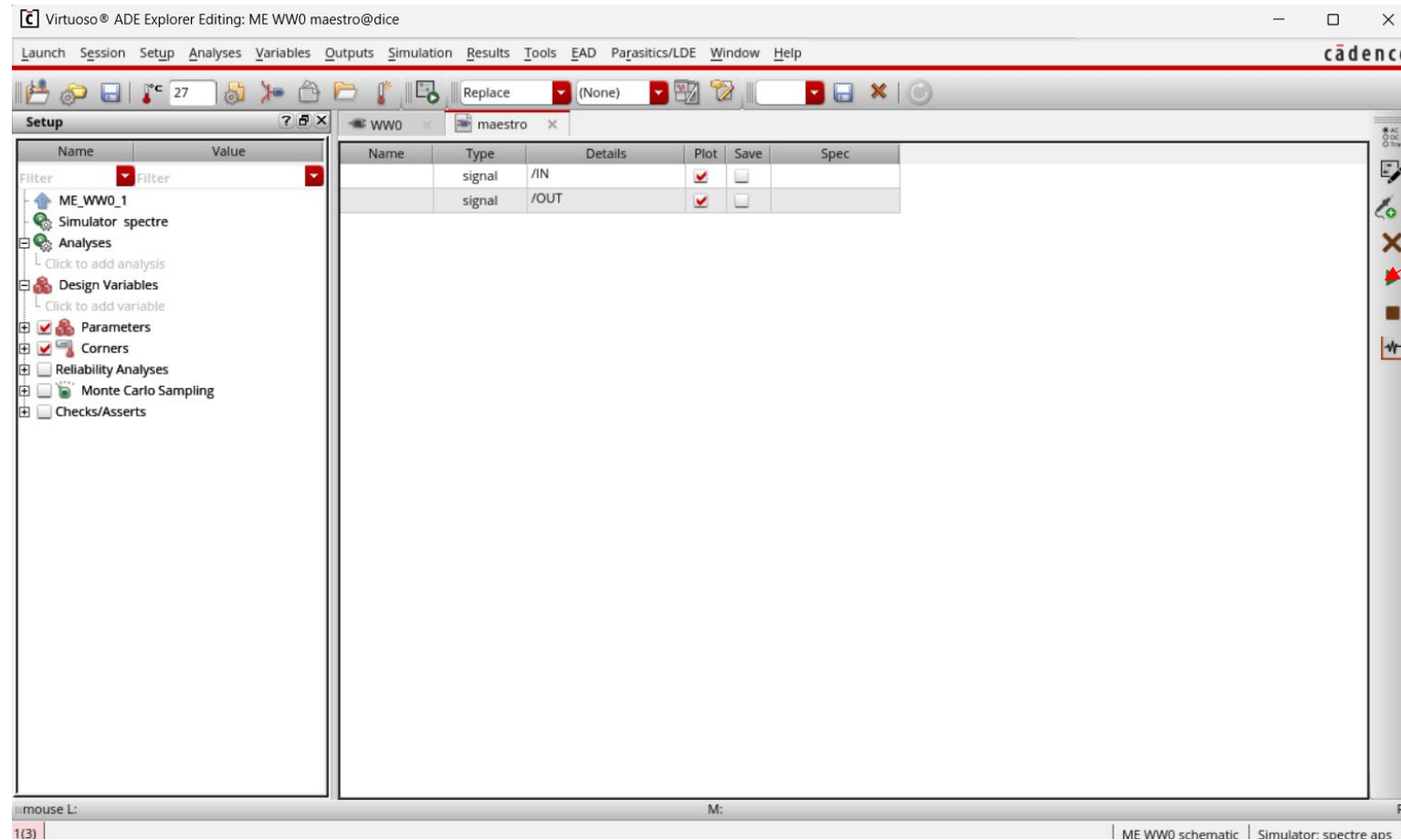
ADE Explorer – probe point selection

Click outputs/To be Plotted / Select On Design

Choose two nets

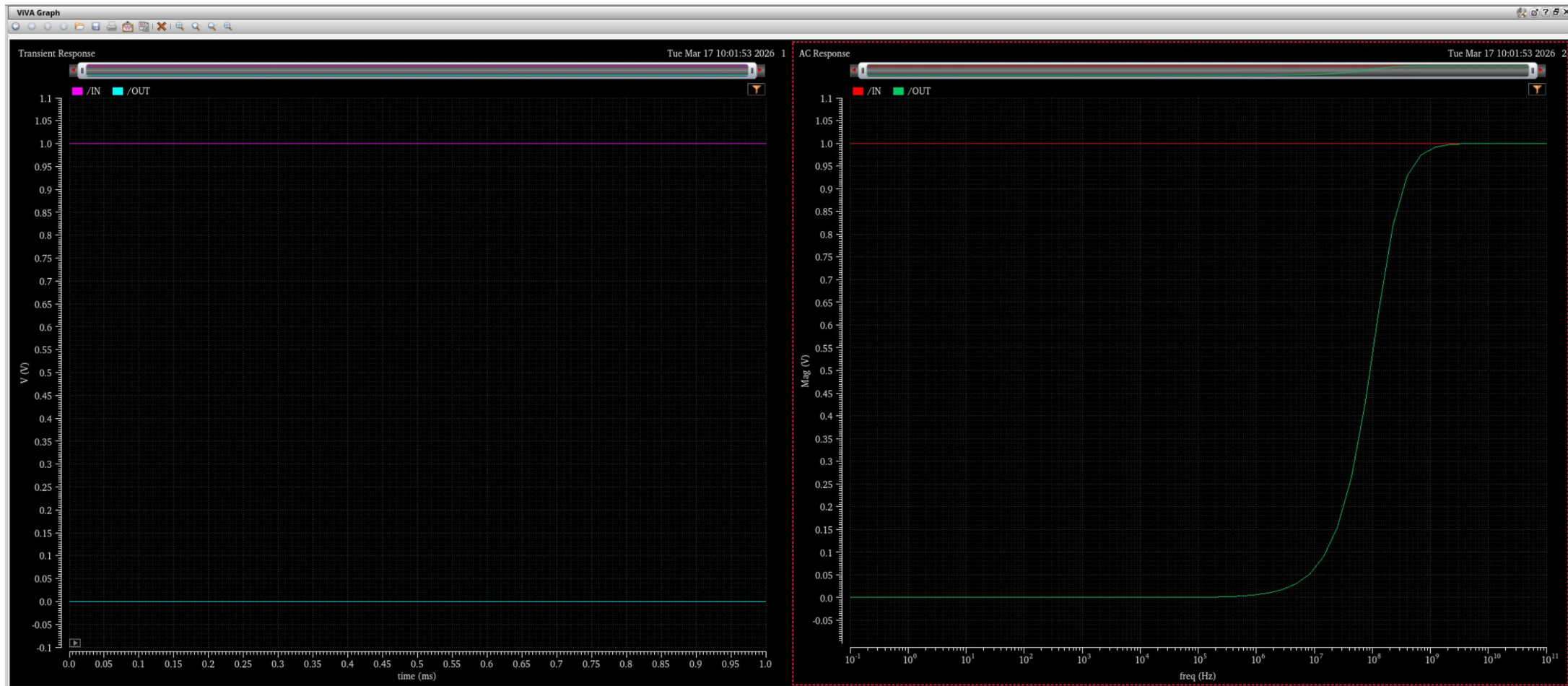
The screenshot displays the Cadence Virtuoso ADE Explorer interface. The 'Outputs' menu is open, showing the 'To Be Plotted' option. A red box highlights the 'Select On Design' option. A red arrow points from the 'Select On Design' option to a circuit diagram. The circuit diagram shows a voltage source (V0) with parameters vdc=1, acm=1, and acp=0, connected to a capacitor (C0) with value 1p. The output of the capacitor is connected to a resistor (R4) with value 1K. The input of the capacitor is labeled 'IN' and the output is labeled 'OUT'. A red box highlights the 'IN' and 'OUT' nodes. A red arrow points from the 'Choose two nets' text to these nodes. The status bar at the bottom indicates '1(3) Select On Design'.

ADE Explorer – run Simulation



Click here

AC Response





1. Select v_f



2. Select dB20

Or type the OCEAN script

The screenshot shows the Cadence Virtuoso ADE Explorer interface. The 'Setup' window is open, displaying a table of simulation parameters. A red box highlights the 'Expression' option in the right-hand menu, with a red arrow pointing to it and the text '3. Click here & select expression'.

Name	Type	Details	Value
	signal	/IN	
	signal	/OUT	
Gain	expr	dB20(VF("/OUT"))	

Name	Type	Details	Value
	signal	/IN	
	signal	/OUT	
Gain	expr	dB20(VF("/OUT"))	
PHASE	expr	phase(VF("/OUT"))	

3. Click here & select expression

- Expression
- Signal
- OCEAN script
- MATLAB expression
- MATLAB script
- Area Specification
- Op Region Spec
- Violation Filter
- OP Parameters

Results

